

LEEUWARDEN AB, Netherlands

WMO: 062700

Lat: 53.223N

Lon: 5.752E

Elev: 1

StdP: 101.31

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-4.9	-10.3	1.6	-5.6	-7.8	1.9	-4.2	14.8	7.7	13.4	7.4	3.8	80	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	27.2	20.2	24.9	19.2	22.9	18.1	21.0	25.8	19.8	23.8	18.8	22.0	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	23.2	18.3	13.2	21.7	17.4	12.5	20.6	60.6	25.8	56.7	23.8	53.3	22.1	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.5	9.3	-10.1	31.4	3.7	1.5	-12.7	32.5	-14.9	33.3	-16.9	34.2	-19.6	35.2		
			-10.4	23.5	3.6	1.1	-13.0	24.2	-15.1	24.9	-17.1	25.5	-19.7	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	2.9	3.2	5.4	8.8	12.1	14.8	17.3	17.3	14.6	10.9	6.8	3.7
	DBStd	6.14	3.90	3.75	3.18	3.51	3.32	2.87	2.85	2.66	2.54	3.11	3.12	3.99
	HDD10.0	958	221	191	144	64	15	0	0	0	1	27	101	196
	HDD18.3	3151	479	423	400	287	196	112	56	52	117	232	345	453
	CDD10.0	906	0	0	2	27	81	146	227	227	137	53	6	1
	CDD18.3	56	0	0	0	0	2	7	23	21	3	0	0	0
	CDH23.3	381	0	0	0	3	17	62	158	122	18	0	0	0
	CDH26.7	83	0	0	0	0	1	13	39	28	2	0	0	0
Wind	WSAvg	4.5	5.3	5.2	5.2	4.5	4.4	4.2	3.9	3.7	4.0	4.4	4.5	4.9
Precipitation	PrecAvg	878	75	60	50	43	59	75	85	95	86	82	85	82
	PrecMax	1098	145	158	99	104	128	141	174	222	193	196	155	176
	PrecMin	610	8	23	9	1	22	27	37	18	15	25	10	36
	PrecStd	132	36	29	27	23	27	33	37	51	42	35	34	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.1	16.4	23.0	25.7	28.6	30.2	29.7	25.9	20.5	15.1	12.3
		MCWB	10.0	9.0	11.2	14.8	17.0	20.7	21.4	20.6	19.2	16.7	13.6	11.3
	2%	DB	9.9	10.1	13.4	19.4	22.7	24.9	27.2	26.7	22.5	17.8	13.6	11.1
		MCWB	8.9	8.6	10.2	13.4	16.3	19.1	20.5	20.1	18.1	15.5	12.5	10.1
	5%	DB	8.8	9.1	11.3	16.8	20.3	22.1	24.7	24.0	20.2	16.2	12.3	10.0
		MCWB	7.9	7.9	9.1	12.3	15.3	17.5	19.6	19.1	16.8	14.5	11.2	9.2
	10%	DB	7.9	8.1	10.0	14.4	18.0	20.0	22.5	22.0	18.7	15.2	11.2	8.8
		MCWB	7.0	7.1	8.3	11.0	14.0	16.3	18.5	18.0	16.0	13.7	10.1	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	10.0	12.4	15.9	19.5	21.4	22.6	22.5	20.2	17.2	13.8	11.7
		MCDB	11.1	11.0	15.0	21.1	23.7	27.1	27.5	27.4	24.5	19.4	14.5	12.2
	2%	WB	9.0	8.9	10.6	14.0	17.2	19.5	21.1	20.7	18.5	15.8	12.7	10.4
		MCDB	9.6	9.8	12.6	18.4	21.1	23.8	26.0	24.8	21.5	17.4	13.5	11.1
	5%	WB	8.0	8.0	9.5	12.7	15.7	18.0	20.0	19.5	17.3	14.8	11.4	9.2
		MCDB	8.6	8.9	11.0	16.1	19.1	21.3	24.0	23.1	19.6	16.1	12.3	9.8
	10%	WB	7.1	7.2	8.4	11.5	14.4	16.7	18.8	18.6	16.4	14.0	10.2	8.1
		MCDB	7.7	8.0	9.8	14.0	17.7	19.4	21.9	21.4	18.3	15.1	11.0	8.7
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.0	6.4	8.1	8.0	7.6	7.7	7.8	7.1	6.1	4.7	4.4
		MCDBR	4.3	5.7	8.8	11.6	11.9	11.5	11.4	10.9	9.2	6.8	4.9	4.2
		MCWBR	4.1	4.7	6.3	7.0	6.9	6.8	6.3	5.8	5.5	5.0	4.4	4.1
	5% WB	MCDWR	4.2	5.1	7.7	10.5	10.6	10.3	10.5	9.8	8.2	6.4	4.9	4.2
		MCWBR	4.1	4.4	5.9	6.7	6.7	6.5	6.2	5.7	5.3	5.0	4.5	4.1
Clear-Sky Solar Irradiance	taub		0.330	0.348	0.376	0.404	0.403	0.401	0.412	0.405	0.388	0.362	0.342	0.322
	taud		2.315	2.339	2.285	2.254	2.313	2.336	2.313	2.347	2.388	2.406	2.346	2.309
	Ebn at Noon		633	746	800	825	847	851	832	816	784	721	612	565
	Edh at Noon		61	81	104	121	120	119	120	110	94	75	59	52
All-Sky Solar Radiation	RadAvg		0.61	1.25	2.57	4.23	5.25	5.48	5.25	4.37	3.05	1.68	0.72	0.46
	RadStd		0.06	0.21	0.25	0.40	0.35	0.36	0.54	0.33	0.27	0.16	0.07	0.05

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-140	+54	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page